

Modeling of Late Melting of Scrap in an EAF

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INTRODUCTION

Late or non-uniform melting of scrap in the Electric Arc Furnace (EAF) is a significant problem in some operations, particularly with large and dense scrap.¹ Late melting may reduce productivity because of delays in tapping the heat, and make chemical sampling inaccurate. Control of this aspect of the **process is hampered** by a lack of understanding of the fundamental melting kinetics in the liquid steel; specifically, how is it affected by scrap size, scrap density, steel temperature, scrap preheating and convection conditions in the steel? The present paper aims at answering some of these questions. **Melting in the liquid heel** is taking on a more important role in EAF steelmaking operations because heel sizes are increasing. The extreme example of this is the CONSTEEL process that operates in flat-bath conditions most of the time.²

The limited amount of work on scrap melting was based on oxygen converters,^{3,4} and used on one-dimensional models to interpret the results. Moreover, the previous work did not address an aspect central to EAF melting; the scrap is very irregular in shape and occupies most of the charge volume. In the present work, a continuum model based on the phase field methodology is used to provide computational solutions for melting of complex pieces of scrap in two dimensions. In particular, the formation of solidified shells around the original scrap pieces, and agglomeration of these shells to so-called “steel icebergs”, usually dominates the melting process. In some EAF operations these “steel icebergs” can be as large as small automobiles.

An additional complication to the process is that there is severe thermal stratification of the steel melt because it is heated from the top and cooled from the bottom, by heat losses. This situation opposes any natural convection, and the liquid steel is virtually stagnant, as will be shown with physical modelling. This stratification is then imposed on the phase field modelling to compute melting times for scrap, which will be shown to be very long.

Bottom gas stirring has been used in some EAFs,^{1, 5-11} but its application is not wide-spread. The benefits in chemical terms have been well-documented; lower oxygen content in the steel, closer approach to slag-metal equilibrium, and lower iron oxide content in the slag, to name a few.¹² The thermal aspects have not been addressed in as much detail; electrical energy savings of the order of 10 to 30 kWh/tonne of steel have been claimed.^{1, 6, 11} The fundamental reasons for such findings will be discussed. Finally, “rules of thumb” will be presented to assist designers and operators to improve melting of scrap in the heel.

MODEL OF STEEL SCRAP MELTING IN THE LIQUID POOL

A phase field-type model of steel scrap melting was recently presented in detail in two previous publications.^{13, 14} It combined an order parameter equation designed to track the motion and interaction of free solid-liquid surfaces with a heat transfer equation which models either diffusion limited conduction in the liquid pool or convective heat transfer using a heat transfer coefficient. The model was used to study the effects of stirring, liquid pool superheat and scrap pre-heating on the melting of single and multiple steel bars. It was also extended to the more complex situations of melting of randomly distributed scrap pieces in the liquid steel. The model was validated against single and two bar experiments.

Phase field modeling of steel scrap melting has revealed that melting time of a scrap assembly in a liquid pool is strongly controlled by the inter-scrap spacing, the initial solid and liquid steel temperatures and the type of heat transfer mechanism in the liquid phase. For situations approximating a perfectly stirred liquid pool at nearly constant liquid steel temperature, simulations revealed that the final melting time was controlled by the melting of largest of a group of isolated “steel icebergs”. Large steel icebergs were found to form predominately in regions of highest scrap density. Figure 1(a-f) shows a typical melting simulation of two-dimensional, randomly distributed, scrap pieces for this situation. Heat is transferred to the solid via convection and the liquid temperature is assumed to remain approximately constant. The color map represents temperature. Figure 1(a) shows the initial temperature distribution and scrap distribution (blue pieces). Soon after melting many individual steel icebergs formed at many locations in the system. The largest ones formed in regions where the local porosity (liquid fraction) was relatively low (Figure 1(b)). The smaller scrap pieces and/or icebergs in high porosity regions were observed to melt quickly and independent of inter-scrap interactions¹³ (Figure 1(a-f)), while the larger ones remained unmelted until late times and, as a consequence, established the final melting time (Figure 1(e-f)).

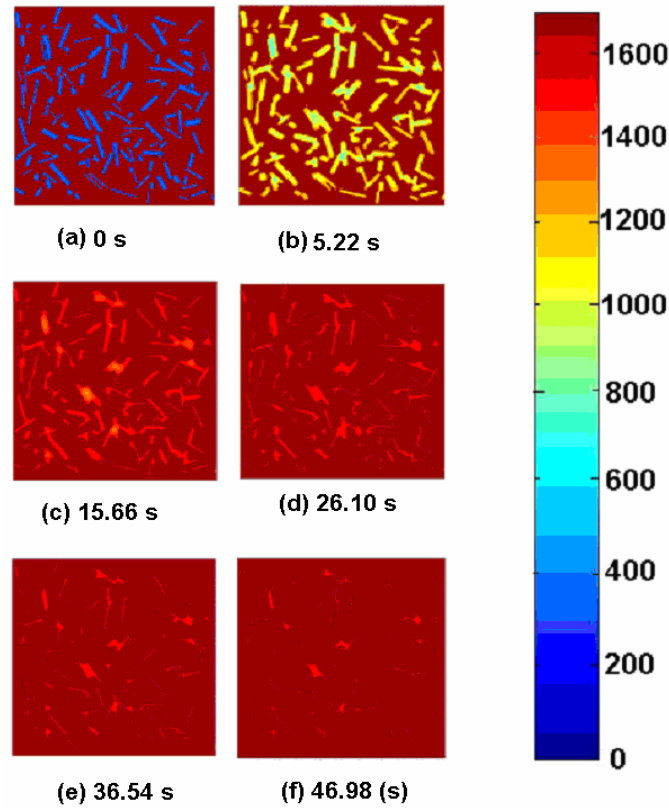


Figure 1. Simulation of a typical melting process of randomly distributed scrap pieces for constant heat transfer coefficient ($5,000 \text{ W/m}^2/\text{K}$). Initial solid fraction is 0.16, similar to bushlings, initial pre-heat temperature $T_o = 25^\circ\text{C}$, liquid pool $T_L = 1600^\circ\text{C}$, melting temperature of steel $T_m = 1520^\circ\text{C}$.

Figure 2 shows simulated melting times as a function of initial solid fraction and heat transfer coefficient for a random scrap assembly similar to that shown in Figure 1. The melting time varies more rapidly with increasing initial solid fraction for a low heat transfer coefficient than it does for a high heat transfer coefficient. In the $5,000 \text{ W/m}^2/\text{K}$ case, as the initial solid fraction increases, there is a rapid rise in melting time in going from partial iceberg formation (occurring in only some areas) to the full iceberg formation (occurring nearly everywhere in the scrap pile). On the other hand, the $13,000 \text{ W/m}^2/\text{K}$ curve is relatively flat, indicating that only partial steel icebergs develop over this range of scrap fractions. These simulation trends suggest that increasing heat transfer coefficient via increased stirring be a very effective way to avoid large steel iceberg formation and correspondingly reduce the final melting time, especially when the scrap density is high.

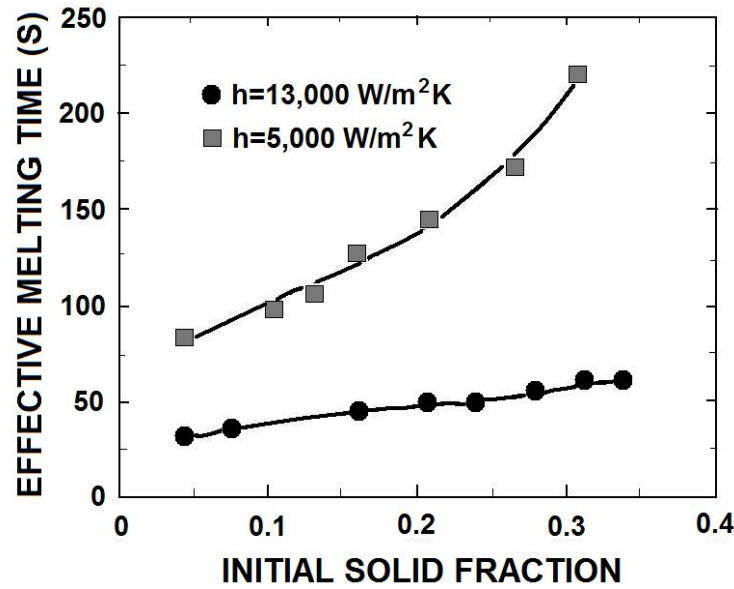


Figure 2. Effective melting time vs. initial solid fraction for different heat transfer coefficients ($T_0 = 25^\circ\text{C}$, $T_L = 1650^\circ\text{C}$, $T_m = 1520^\circ\text{C}$ solid lines are regression fits to the points.)

The modelling results are consistent with experimental observations. Figure 3 illustrates the relationship between porosity and the melting time from experiments where an assembly of multiple, parallel bars was melted in an induction furnace.¹⁴ At high effective porosity (porosity is defined through the distance between bars) the bars melt independently of the other, and the final melting time increases only slightly as porosity decreases. With further decrease in porosity, the solidified shells that grow around pieces close the center of the multi-bar sample assembly begin to touch each other. This is because the superheat and the heat transfer coefficient are relatively low in the central area of the scrap assembly. As expected from simulations such as those of Figures 1 and 2, such a decrease in the distance between the bars leads to partial steel iceberg formation and a rise in the final melting time of the multi-bar assembly. As bar spacing decreases further, there is a sharp increase to almost complete steel iceberg formation throughout the scrap assembly, and a corresponding sharp rise in the final melting time.

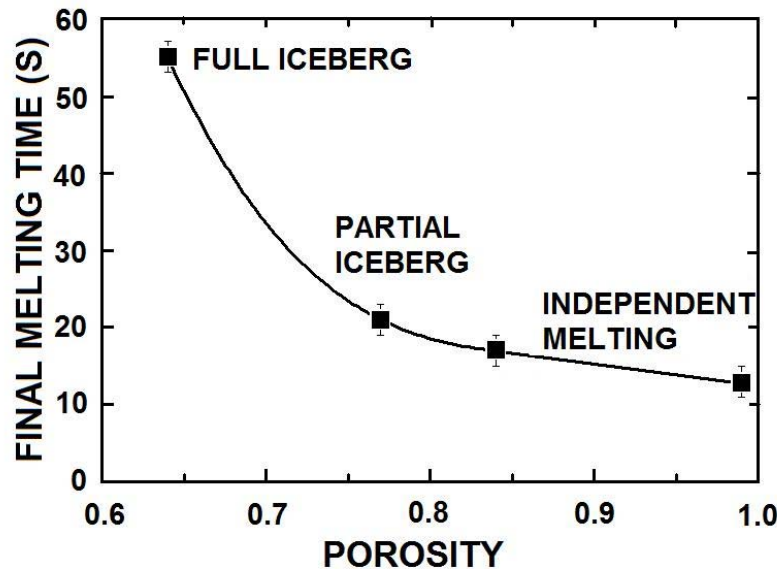


Figure 3. Final melting time vs. porosity for complex multi-bar sample determined experimentally. See Ref. 14 for details.

The phase field modeling approach can also be used to better understand the role of superheat in the pool and the effect of pre-heating the scrap on the final melting time. Figures 4 and 5 show the melting times as a function of porosity for different liquid pool and scrap pre-heating temperatures, respectively. It was also shown that the melting time increased as the size of the scrap increased.¹⁵

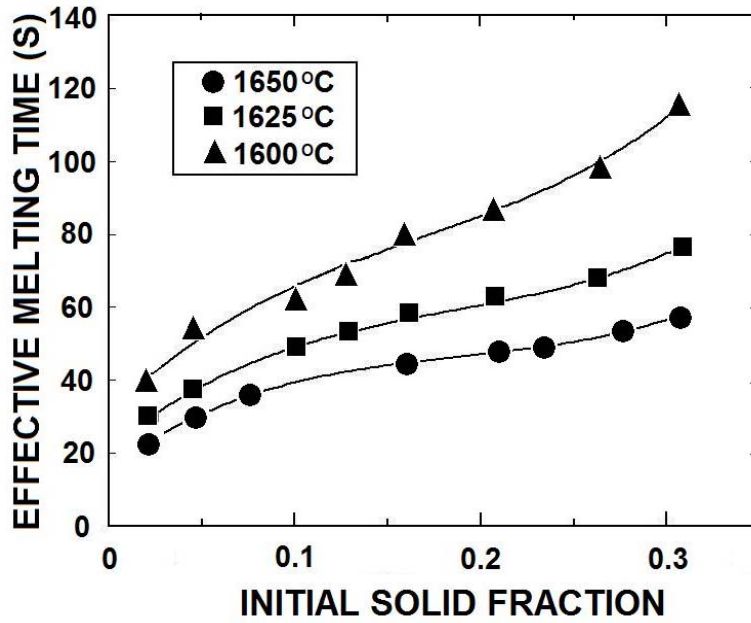


Figure 4. Effective final melting time vs. initial solid fraction for various liquid steel temperatures ($T_0 = 25^\circ\text{C}$, $T_m = 1520^\circ\text{C}$, $h = 13000 \text{ W/m}^2/\text{K}$. Solid lines are regression fits to the points)

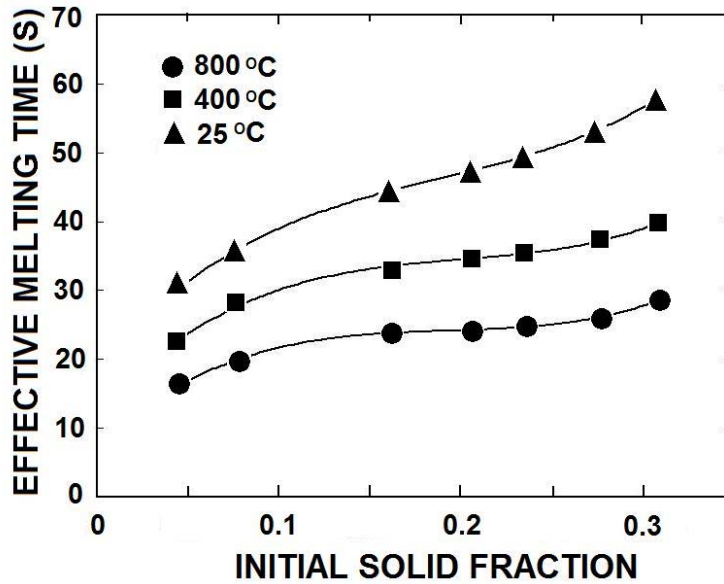


Figure 5. Effective melting time vs. initial solid fraction for various initial steel scrap pre-heat temperatures ($T_L = 1650^\circ\text{C}$, $T_m = 1520^\circ\text{C}$, $h = 13000 \text{ W/m}^2/\text{K}$, solid lines are regression fits to the points.)

MODELLING CONVECTION IN EAFS

The preceding section highlights the importance of convection-dominated stirring in the bath for melting steel scrap; in this section convection will be addressed. Several years ago one of the authors investigated thermal stratification in submerged arc electric furnaces.²² While submerged arc electric furnaces are used for smelting nickel and precious metal feed materials, they share several features with electric arc furnaces for steelmaking:

1. They are relatively shallow and wide, usually with an arched bottom.
2. They are heated from the top and cooled from the bottom, which opposes natural convection.
3. They are traditionally unstirred.

Therefore, it is useful to interpret this work for EAFs.

A low-temperature 1-/10-scale model was constructed of a portion of the Falconbridge furnace (now Xstrata Nickel). An aqueous solution of calcium chloride was used to simulate the metallic matte phase, while oil was used to simulate the slag. Heated steel cans, to simulate the electrodes, supplied heat from the top, and water cooling of the container simulated heat losses from the bottom of the furnace. The model was designed to ensure reasonable dimensional similarity with the real system.

Figure 6 shows that the steady-state temperature profiles in the experimental model were linear. The product of the gradient and the thermal conductivity in each phase was constant, suggesting that heat is flowing vertically downward through stagnant fluids. The heat balance for the furnace confirmed that the heat loss through the bottom could be calculated with Fourier's Law:

$$Q = -kA \frac{dT}{dx} \quad (1)$$

Therefore, heat is conducted, not convected, through the hearth because heating from the top and cooling from the bottom makes the system stagnant.

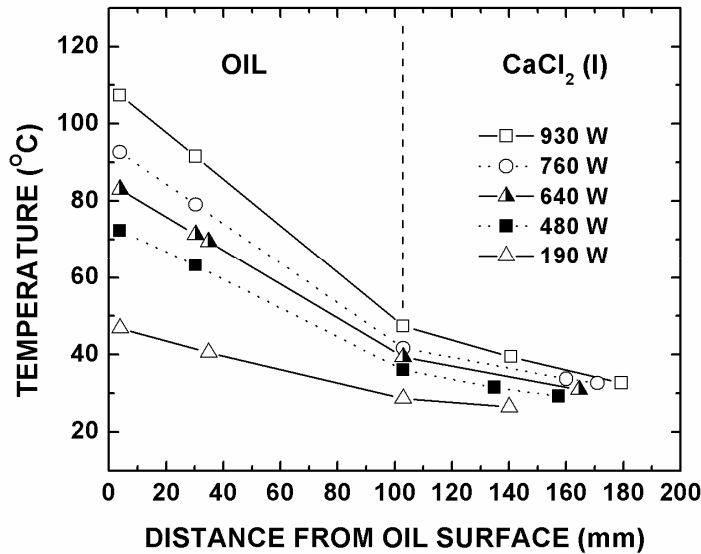


Figure 6. Temperature profiles in the unstirred experimental furnace model at various power levels.

The study then went on to examine the possibility of eliminating the stratification with gas stirring. Figure 7 shows how four thermocouples in the model respond to gas stirring. They start with the stratification observed in Figure 6; the stratification is reduced, but not eliminated, even at this relatively high flow rate.

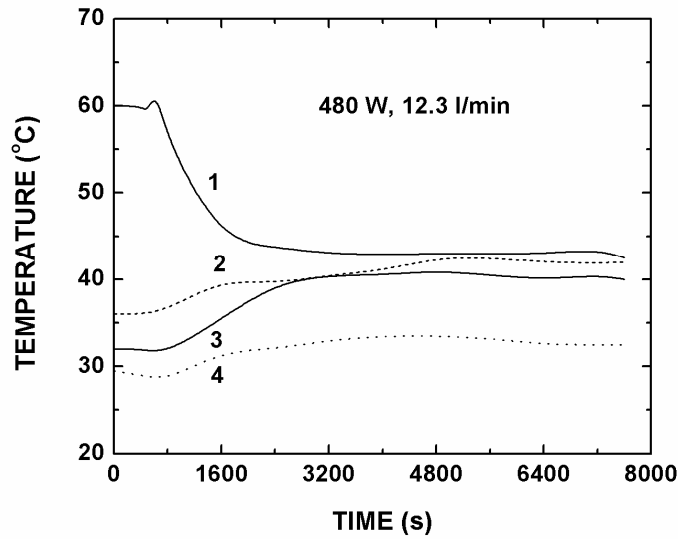


Figure 7. The progress of thermal homogenization for electrical power input of 480 W and gas injection rate of 12.3 l/min (through 6 injectors). Thermocouples 1 to 4 were positioned from top to bottom in the fluids.

Figure 8 summarizes the steady-state gradients with various levels of gas stirring. It is clear that gas stirring cannot completely homogenize a melt that has an imposed heat source from the top. It was also shown that the mixing times were significantly influenced by the heating. Therefore, isothermal water models of EAFs cannot be used to assess mixing phenomena.

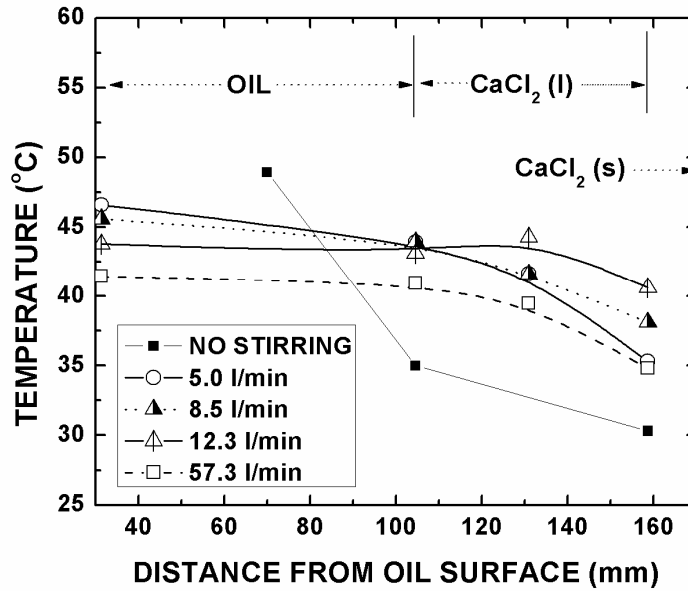


Figure 8. The steady-state thermal profiles for various gas injection flow rates, compared to the unstirred case, with 480 W of electrical power.

SCRAP MELTING WITH IMPOSED STRATIFICATION

In light of the previous results, the phase field melting model¹⁴ is now applied to case where there is an imposed top surface temperature to simulate scrap melting with thermal stratification. It was convenient to specify an effective value (k_{eff}^L) which could

be the liquid conductivity, or could be larger by some enhancement factor to investigate the effects of convection. From the foregoing discussion the enhancement factor is expected to be close to unity, and furthermore, the extent of convection between the pieces of the scrap pile is expected to be small.

Figure 9 shows a typical simulation of randomly distributed scrap pieces for this case. The scrap pieces (blue) were uniformly distributed in the liquid steel (red) initially. Starting from the early stage of simulation, large steel icebergs were widely distributed throughout the liquid, in contrast to the small, isolated “islands” present in the case of convection-dominated melting simulations shown in Figure 1. The temperature of the scrap reached the melting temperature in the very early stages of melting. This implies that the sensible heat contained in the liquid steel was very rapidly consumed for increasing the steel scrap temperature to the melting temperature.

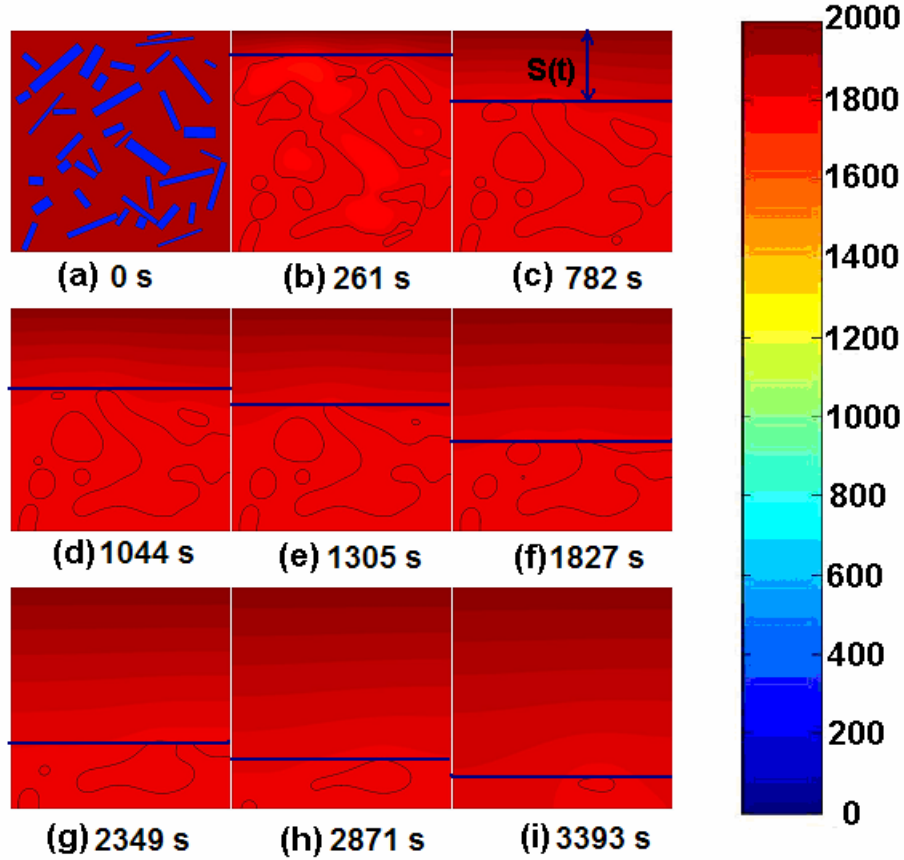


Figure 9. Typical melting process of randomly distributed scrap pieces for simulation with k_{eff}^L (Initial Solid Fraction = 0.167, $T_0 = 25^\circ\text{C}$, $T_L = 1650^\circ\text{C}$, $T_m = 1520^\circ\text{C}$, $k_{eff}^L = 4k_L$). The time of each frame is indicated. The initial scrap distribution is shown in frame (a). The top boundary temperature was 200°C higher than the melting temperature of steel scrap.

Analysis of simulations such as those in Figure 9 reveals that the melting front advances from top to bottom through an effective one-dimensional melting front line (horizontal blue lines in Figure 9), defined as the demarcation line between the region of complete liquid (above the melting point) from the region containing a solid/liquid mixture (very close to the melting point). The distance of the blue line from the top surface to the melting front line, denoted $S(t)$, is well approximated by

$$S(t) = \frac{2\gamma}{\sqrt{\rho C_p}} \cdot \sqrt{k_{eff}^L} \cdot \frac{1}{\sqrt{ISF}} \cdot \sqrt{t} \quad (2)$$

where γ is given by the solution of $\gamma \cdot e^{\gamma^2} \operatorname{erf}(\gamma) = (T_{top} - T_m) \frac{C_p}{\Delta H_f \sqrt{\pi}}$, T_{top} and T_m are the top boundary and melting temperatures, respectively and ISF represents *I*ns*t*an*e*ous *S*olid *F*raction along the melting front line. This can be estimated by

$$ISF \approx V_s - \frac{C_p(T_L - T_m)}{\Delta H_f}(1 - V_s) + \left(\frac{C_p}{\Delta H_f}(T_m - T_0) \cdot V_s \right) \quad (3)$$

where V_s is the initial solid fraction in the liquid pool. Figure 10 shows a plot of the dimensionless melting front position, against dimensionless time $\bar{t} = t/(0.4^2/P^2)$, where 0.4 m is the size of simulation system and P is the expression on the right hand side of Eq. (1) that is multiplying the $\sqrt{t}$.

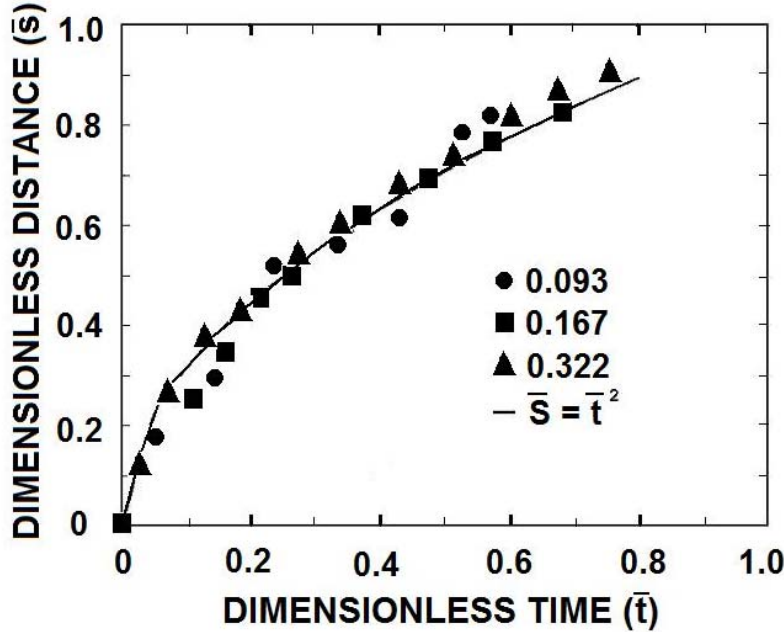


Figure 10. Scaling collapse of simulated melting front position vs. time for various initial solid fractions onto a parabolic curve using the one-dimensional analytical model ($T_0 = 400^\circ\text{C}$, $T_L = 1600^\circ\text{C}$, $k_{eff}^L = 4k_L$)

It should be noted that equation 2 can only be used to describe the heat conduction-dominated melting process under the condition that the temperature of the solid-liquid mixture region reaches approximately the melting temperature at the very early stages of melting. This condition requires that the characteristic time for reaching the melting time in the solid/liquid mixture region is much smaller than the characteristic time of the melting front sweeping through the system set by equation 2. Outside these parameter ranges this simple effective 1D analytical solution is no longer valid and the phase field model must be used to describe the melting process.

IMPLICATIONS FOR SCRAP MELTING IN EAFS

Most EAFs do not have bottom stirring, so that the melting of scrap in the liquid steel may resemble Figure 9. This is a very slow process. Liquid steel that has been melted by the arcs will have some superheat, and this superheat must be conducted, not convected, to the scrap below because the melt is so stagnant. Figure 9 illustrates this point very well; for a melt depth of 0.4 m, a steel superheat temperature, T_{top} , 1720°C , initial steel temperature of 1650°C , a melting point of 1520°C , an initial solid fraction of 0.16 (similar to bushelings), the time for 80% melting is 3393 s (57 minutes). Therefore, it is difficult to ensure complete melting of scrap during normal heat times. Contrast this with Figure 1 where convection in the liquid steel of a constant temperature completes the melting in less than 1 minute. These two figures show the two extremes of melting behavior; in reality, it may be somewhere in between, depending on the extent of convection. Nevertheless, it is useful to examine these two extremes.

Equation 2 can be used to provide useful “rules of thumb” for melting steel scrap in unstirred liquid steel. It shows that the time for melting is proportional to the initial solid fraction, so denser scrap will take longer to melt. Equation 2 also shows that the time for melting increases with the square of the superheat, so this is a very important operational variable. It also shows that for small enough scrap pieces the size of the scrap is not very important; this is because the diffusion of heat from the top controls the process, not diffusion of heat into each piece of scrap. Equation 2 also shows that the time goes up with the square of the distance of melting front line. For example, if the liquid steel depth is increased by 25%, then it will take 56% longer to melt. There is a trend to larger heel size; Siemens is recommending for their latest furnaces (SIMETAL CIS ULTIMATE) that heel size be 25% of the tap weight.²³ The

extreme case of large heel size is the CONSTEEL process, which operates in flat bath conditions most of the time. If one wants to take advantage of larger heel sizes to melt scrap more quickly, it is important to move away from this regime of heat diffusing from the top to a regime where the liquid steel temperature is uniform.

There are three ways that liquid steel can be effectively stirred in an EAF to produce a more uniform liquid steel temperature:

1. Bottom gas injection.
2. Gas evolution from chemical reactions.
3. Electro-magnetic stirring.

Each will be discussed in turn.

Previous modelling work of stirring in EAFs has shown that bottom stirring is far more effective than top lance, forces from the arcs or slag foaming in stirring liquid steel.²⁴⁻²⁶ The work described in Section 3 shows that gas stirring can reduce, but not eliminate temperature gradients in furnaces that are heated from the top.

As mentioned earlier, several companies have reported that bottom-stirring saves 10 to 30 kWh/tonne of steel.^{1, 5-11} Our interpretation is that the hot liquid steel from the top is brought into contact with the scrap, to melt the scrap in a shorter time. There may be other benefits of reduced heat losses due to a more uniform furnace temperature, and gains in productivity.

The main reaction that can be used for gas evolution is the carbon boil. One of the major advantages of pig iron as an alternative iron source is the carbon boil that can be created during its melting. However, the timing and duration of the boil may not match the need for stirring to assist scrap melting. DRI and HBI also create carbon boils, but if they are top charged they may not provide deep stirring.²⁷

Electro-magnetic stirring has not been installed in EAFs for over 20 years, but could be useful for high productivity shops with larger heel sizes.²⁸

If one or more of these techniques is used, then the scrap will melt much more quickly, as illustrated in Figure 1. The characteristics of the melting are much different than in the regime where melting is controlled by conduction from the top.

Figure 2 shows that intense stirring, as reflected in the higher heat transfer coefficient, has a dramatic effect on melting time. For the lower heat transfer coefficient in Figure 2, there was significant steel iceberg formation at the higher solid fractions. This highlights a dilemma for melting denser scrap in the heel; the computations are based on a uniform heat transfer coefficient for each piece of scrap, yet it may not be possible to achieve this in reality for pieces that are in deep in the scrap charge. Therefore, there may still be significant steel iceberg formation for dense scrap.

Figure 3 presents a useful “rule of thumb” for the propensity for steel iceberg formation and its effect on melting time. For scrap of high porosity, such as shredded scrap, ~0.9, there is little bridging between scrap pieces, but as one moves to bushings, and into compressed bundles, there is much more of a penalty in melting speed, up to a factor of 5.

Figure 4 shows that the steel superheat has a significant effect on melting times. The melting point of the steel for the computations was 1520°C, so the superheat ranges from 80 to 130°C. One of the features of ultra-high power (UHP) furnace operation is that the steel melted by the arcs will have significant superheat, although it is difficult to calculate how much. The liquid will trickle down to the heel, so it is important that it be convected through the heel, and not remain stratified. This is another way to explain the benefits in melting observed with gas stirring.

Preheating of the scrap also has a significant effect as shown in Figure 5, and it is one of the key benefits of the CONSTEEL process. It should be noted that these computations are based on effective convection, but most CONSTEEL operations are not stirred. TECHINT recommends bottom stirring for all their CONSTEEL design. Some customers decide not to use it to reduce Argon consumption and because of concerns about maintenance and bottom wear, which in reality are not significant. The latest CONSTEEL designs in Europe (Acciaierie Arvedi (250 tonnes) and Trier (60 tonnes)) are equipped with bottom stirring.²³

CONCLUSIONS

1. The process of scrap melting in liquid steel is dominated by the formation of solidified shells of steel around the original scrap pieces. The agglomeration of these shells into steel icebergs delays the melting of the scrap, depending on specific conditions.
2. Because EAFs are heated from the top, and cooled from the bottom, they are stably stratified, resulting in near stagnant conditions in the liquid steel.
3. In this stably stratified condition, heat must diffuse downward to melt scrap in the heel. This is a slow process of the order of 1 hour. From the modelling, the following “rules of thumb” were developed for this regime regarding the time for scrap melting; it
 - is proportional to the scrap density,
 - increases with the square of the heel depth,
 - decreases with the square of the superheat, and
 - is independent of the scrap size for sufficiently small scrap pieces.
4. To move out of the stratified regime, stirring is required. Scrap melting is much faster (a matter of minutes in the simulations) in a convection-dominated regime. In this regime, the time for scrap melting:
 - increases with convection (higher heat transfer coefficient),
 - increases dramatically as scrap density increases because of greater steel iceberg formation,
 - increases with scrap size,

- decreases with steel superheat,
 - decreases with scrap preheat, and
 - is independent of the heel size, in so far as the convection is uniform throughout.
5. The specific energy requirements for melting (kWh/tonne) are reduced with gas stirring because the stratification is reduced, and heat from the top is brought into contact with the scrap by convection.
 6. There is a trend to make the heel size larger in EAF to accelerate scrap melting. To take full advantage of this, stirring will help.
 7. Isothermal water models of EAFs give misleading estimates of mixing and flow patterns because the thermal stratification suppresses the convection significantly.

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